ATKINS

CABLESIZER VERSION – 1.1

USER MANUAL

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1. QUICK GUIDE TO THE USER

1.1 PREFACE

CableSizer Version-1.1(hereafter mentioned as CS) is the first release of the spreadsheet customized to carry out various calculations pertaining to sizing of feeder, preparing DB load schedules, SMDB load schedules, MCC load schedules, Busriser load schedule, MDB load schedule & Transformer load schedule in a much shorter time. This manual is aimed at providing guidance to the users about using CS. The objective of section-1 is to give a broad idea about CS and to enable the user to start working with it immediately. The subsequent sections of this manual will cover a detailed interpretations of the various formulae used in the CS.

The back-end of CS is written in Visual Basic (VB) using very simple equations. The back-end is used only as a database to store various cable/busbar parameters. It just performs various checks on the inputs given and gives the data sought whereas the actual calculations are being in the front-end (MS Excel) alone. Throughout the CS user will encounter various user-defined functions apart from the in-built library functions. These functions are used to look in the back-end for the data required for performing the calculations.

All the results will be generated in the front-end only and if the user is not sure about the results generated it is then recommended to contact the developer or if the user is thorough about the working of CS he/she can manipulate/troubleshoot on his/her own. It is strongly recommended to not to change any formula or the VB codes without consulting the developer.

1.2 ADOPTED DESIGN STANDARDS

The CS is very much based on the IEE wiring regulations 16th Edition (BS 7671 : 2001). However, few references have been made to IEC-781 (for 3phase symmetrical short-circuit calculation) and IEC-364 regarding the installation methods and grouping factors. Few references have been made to IEE guidance notes-5 & 6 and ERA 840067 in the calculation of Maximum Earth Fault Loop Impedance. The CS also takes care of the minimum requirements laid down by the DEWA regulations on the sizing of cables.

1.3 LOOK-UP TABLES

The current carrying capacities, derating factors except grouping factor for trays have been taken from the Ducab catalogue. The grouping factors for installation on trays have been taken from IEC 364-5-523. The voltage drop values have been taken from the IEE wiring regulations – Appendix-4. The resistance values for armour @ 20°C have been taken from BS 5467:1997 & BS 6346:1997.

The resistance & reactance values for Busbars have been taken from Power Bar Catalogue to be more conservative than other Busbar data.

1.4 INPUT & OUTPUT CONVENTIONS

As a general rule all the cells highlighted in yellow requires manual input from the user. Cells highlighted in cyan indicate the value is generated automatically but still it requires the user to verify it and may modify the value depending on the check remarks or as required by the user. All other colors are intended to add some cosmetics to the sheet.

Warning: Any cells over-written by the user shall be colored in yellow so as to notify the user of the manual entry in future when the user or someone else looks at it and also in case of any modification to any input the cell will not reflect automatically of the input changes!!!!

Do not insert or delete any column in any of the sheets. It may result in the collapse of the whole calculation.

Any font appearing in red color indicates a warning and requires user's attention. In such cases refer to the check remarks appearing at the last column of the sheet. Fonts appearing in blue color don't indicate any warning. It is intended to notify of some harmless deviation from any regulations.

Note: Cells highlighted in green shall not be changed under any circumstance unless they are found to display any wrong values quoted in the regulation it is based on.

Although the section-1 provides a quick guide for the user to start with using CS, it is strongly recommended to read all the sections of this manual before actually using it.

1.5 Disclaimer

CS Version 1.1 is intended to provide only a design guideline to the user. Several cases have been tested with CS and the results are found to be inline with the manual calculations. Although, the results are to a large extent in agreement with the various internationally accepted standards, it shall be acknowledged that the CS is not fully automated nor does it incorporate any artificial intelligence to decide on its own. It is the sole responsibility of the user to verify & rely on the results generated. The developer or the approver is not responsible for any inaccuracies or damage resulting from its use.

The back-end of CS is the intellectual property of the developer and W.S.Atkins & Partners Overseas (WSA) and so is the information contained in this manual. CS is approved for use by the Staff of WSA alone.

2 - LIST OF SYMBOLS

P	Active Demand Load in kW
Q	Reactive Power Rating in kVAR
I_b	Demand current in A
V_n	Nominal System Voltage in V
I_n	Overload Current Setting of Protective Device in A
n	Magnetic Current Setting Multiplier
I_m	Magnetic Current Threshold in A
S_m	Apparent Motor Load in kVA
I_L	Locked Rotor Current for Motor in A
I_F	Full Load Current for Motor in A
Z'_m	Transient Impedance of Motor in mΩ
R'_m	Transient Resistance of Motor in mΩ
X'_m	Transient Reactance of Motor in mΩ
K_1	Temperature Derating Factor
K_2	Grouping Factor
Kr	Soil Thermal Resistivity Factor
K_3	Depth of Laying Factor
KTOT	Overall Derating Factor
L	Cable/Busbar Length
I_t	Nominal Current Rating of Cable in A
I_z	Corrected Current Rating of Cable in A
VDr	Resistive Voltage Drop in mV/A/m or mΩ/m
VDx	Reactive Voltage Drop in mV/A/m or mΩ/m

LIST OF SYMBOLS (Continued)

VDz	Unit Voltage Drop in mV/A/m or mΩ/m
Vc	Circuit Voltage Drop in V
Vup	Upstream Voltage Drop in V
Vt	Total / Cumulative Voltage Drop in V
S_{up}	Upstream fault level in MVA
S	Apparent Power Rating of Transformer/Generator in kVA
Uk	Percentage Impedance of the Transformer
Ur	Percentage Resistance of the Transformer
Ux	Percentage Reactance of the Transformer
ρ	Resitivity of the conductor in mΩ.m
R_k	Transformer/Conductor Resistance at 20°C in mΩ
X_k	Transformer/Conductor Reactance at 20°C in mΩ
R_{kt}	Total/Cumulative Conductor Resistance at 20°C in mΩ
X_{kt}	Total/Cumulative Conductor Reactance at 20°C in mΩ
Z_{kt}	Total/Cumulative Conductor Impedance at 20°C in mΩ
Isc	Prospective 3Phase Symmetrical Short Circuit Current in kA
k1	Factor accounting for Resistivity, Temperature Coefficient, Heat Capacity and Initial & Final Temperatures for the Phase Conductor
t1	Disconnection Time in s for the Protective Device for Isc
S_1	Cross-sectional Area of Phase Conductor in mm²
X"d	Percentage Sub-transient reactance of Generator
Y	Admittance in S
G	Conductance in S
B	Susceptance in S

LIST OF SYMBOLS (Continued)

Y_m	Admittance of Motor in S
G_m	Conductance of Motor in S
B_m	Susceptance of Motor in S
Y_k	Admittance of Tranformer/Cable in S
G_k	Conductance of Tranformer/Cable in S
B_k	Susceptance of Transformer/Cable in S
Zup	Upstream Earth Fault Loop Impedance at 20°C in mΩ
Rc	Phase Conductor Resistance at 20°C in mΩ/m (contributing to Earth Fault Loop Impedance)
Recc	Earth Continuity Conductor Resistance at 20°C in mΩ/m (contributing to Earth Fault Loop Impedance)
Ra	Armour Resistance at 20°C in mΩ/m (contributing to Earth Fault Loop Impedance)
Rp	Combined Resistance of the Protective Conductor Comprising ECC & Armour (as applicable) at 20°C in mΩ/m (contributing to Earth Fault Loop Impedance)
Xc + Xp	Phase – Protective Conductor Reactance Loop at 20°C in mΩ/m (contributing to Earth Fault Loop Impedance)
Zu	Unit Earth Fault Loop Impedance at 20°C in mΩ/m
Zc	Circuit Earth Fault Loop Impedance at 20°C in mΩ
Zt	Total/Cumulative Earth Fault Loop Impedance at 20°C in mΩ
Ief	Maximum Earth Fault Loop Current in kA
k2	Factor accounting for Resistivity, Temperature Coefficient, Heat Capacity and Initial & Final Temperatures for the ECC
t2	Disconnection Time in s for the Protective Device for Ief
S_2	Cross-sectional Area of ECC in mm²
Z'up	Upstream Earth Fault Loop Impedance at Fault Temperature in mΩ
R'c	Phase Conductor Resistance at Fault Temperature in mΩ/m (contributing to Earth Fault Loop Impedance)
R'ecc	Earth Continuity Conductor Resistance at Fault Temperature in mΩ/m (contributing to Earth Fault Loop Impedance)

LIST OF SYMBOLS (Continued)

R'a	Armour Resistance at Fault Temperature in mΩ/m (contributing to Earth Fault Loop Impedance)
R'p	Combined Resistance of the Protective Conductor Comprising ECC & Armour (as applicable) at Fault Temperature in mΩ/m (contributing to Earth Fault Loop Impedance)
X'c + X'p	Phase – Protective Conductor Reactance Loop at Fault Temperature in mΩ/m (contributing to Earth Fault Loop Impedance)
Z'u	Unit Earth Fault Loop Impedance at Fault Temperature in mΩ/m
Z'c	Circuit Earth Fault Loop Impedance at Fault Temperature in mΩ
Z't	Total/Cumulative Earth Fault Loop Impedance at Fault Temperature in mΩ
Zm	Maximum Allowable Earth Fault Loop Impedance in mΩ

3. CALCULATION FUNDAMENTALS

Under this section the user will find the formulae being used in CS, the interpretation of the associated standards/formulae and various assumptions made. It is intended to provide a simplified approach to the Electrical calculations being done using CS. It is strongly recommended to get familiarized with all the concepts illustrated throughout the section prior to using the CS.

CABLE SIZING CALCULATIONS:

In order to select a cable for a particular application, it is deemed necessary to carry out the following calculations as a minimum:

- Current Carrying Capacity of the Cable
- Voltage Drop
- Prospective 3Phase Symmetrical Short Circuit Current
- Prospective Earth Fault Current
- Let Through Energy or Thermal Constraint
- Maximum Earth Fault Loop Impedance

The above calculations will be dealt in an elaborate manner in the forthcoming sections.

3.1 CURRENT CARRYING CAPACITY:

The nominal current carrying capacity of a cable at 30°C can be obtained directly from the cable manufacturer's catalogue. The above rating is then subjected to the following derating factors depending on the type of installations:

- Ambient temperature derating factor K_1
- Grouping Factor K_2
- Soil Thermal Resistivity Factor Kr
- Depth of Laying Factor K_3

The overall derating factor is given by $K_{TOT} = K_1 \times K_2 \times Kr \times K_3$

The corrected current capacity of the selected cable $Iz = It \times K_{TOT}$

For a cable to be selected based on its current carrying capacity

$$\mathbf{Iz > In}$$

Where In is the overload current setting of the protective device.

Why should I consider derating factors?

The tabulated current capacities in the catalogue are measured/calculated under a preset condition which is different from the actual site conditions. 4

The values of the derating factors and nominal current carrying capacity of the cable can be obtained from the cable manufacturer's catalogue.

For ambient temperature derating factor, the value corresponding to a temperature of 50°C shall be considered for installation in Air and 35°C for underground installations. The ambient temperatures may vary from region to region. The 50°C and 35°C are applicable to UAE.

For grouping factor, it is not always wise to consider all the cables/circuits adjacent to the subject cable for grouping. For instance, if the subject cable is anticipated to carry a current of the order of 1000A or so and is run alongwith a cable anticipated to carry just 5 or 10A then it would have negligible effect on the temperature rise of the subject cable and hence need not to be considered for grouping. Similarly, a cable installed at one end of a wide cable tray will have negligible effect on the cable on the other end and hence need not to be considered. Therefore due considerations shall be given in choosing the appropriate number of circuits/cables for grouping in order to avoid too pessimistic and an uneconomical cable selection.

Soil thermal resistivity is a measure of increase in the temperature of soil adjacent to a metre length of cable per Watts of power dissipated across the length of the cable. The tabulated values relate to a resistivity value of approximately 1.2 °Cm/W. This value varies from region to region. A typical value of 2.0°Cm/W is applicable for UAE. For any other values of Soil Thermal Resistivity the derating factor applies.

In general, the temperature rise of a cable is limited as much as it is closer to the air which implies that a cable laid deeper will be derated to a more extent than a shallow one in order to compensate for the temperature rise. The tabulated values relate to a laying depth of 0.5m. For any other depths, the derating factors apply.

3.2 VOLTAGE DROP CALCULATION:

The maximum allowable voltage drop as per IEE wiring regulation clause 525-01-02 is 4% starting from the source until the last point of the farthest circuit. The only exceptions to this criterion are the voltage appearing at any motor terminals during starting, equipment with inherent starting current inrush characteristics. For such exceptions the voltage drop shall not exceed the acceptable tolerance given in the relevant British Standards or as recommended by the equipment manufacturer.

DEWA regulation allows a voltage drop of 4% from the point of origin of supply till the farthest point of any circuit.

The general formula for calculating the voltage drop encountered in a cable is given below:

$$Vc = \frac{Vu\ (\text{Tabulated mV/A/m}) \times Ib \times L}{1000} \quad V$$

The above formula holds good when the phase angle of the cable equals the phase angle of the load otherwise it will result in a pessimistically higher value of voltage drop. The Clause -7.2 of Appendix-4 of IEE Wiring Regulations gives a more appropriate formula for calculating the voltage drop considering the load power factor.

$$Vu = VDr \cos\Phi + VDx \sin\Phi \quad mV/A/m$$

$$Vc = \frac{Vu \times Ib \times L}{1000} \quad V$$

$$Vt = Vc + Vup \quad V$$

$$\%Vt = \frac{Vt \times 100}{Vn} \quad \%$$

The voltage drop values for the cables of various types and various installation methods for single phase and three phase circuits are given in the Appendix-4 of IEE Wiring Regulations. The resistive and reactive voltage drop values are given for cable sizes of 25mm² and above. Smaller cable sizes have negligible reactive voltage drop and hence only unit voltage drops have been furnished.

It is worth noticing in the table that the values given do not relate to a single conductor in both single and three phase application. Rather, it relates to single phase or three phase **circuit** unit voltage drop. Consequently, the single phase voltage drop value divided by 2 and the corresponding three phase circuit value divided by √3 shall be equal. Further all the values furnished in the tables relate to the cable operating at its maximum operating temperature (90°C for XLPE and 70°C for PVC). If the user intends to dig much on the actual operating temperature of the cable and the corresponding unit voltage drop value, Appendix-4 of IEE Wiring Regulations shall be referred.

3.3 SHORT CIRCUIT CURRENT CALCULATION:

The short circuit calculation being referred in this section implies the prospective 3phase symmetrical short circuit current at a given point arising due a bolted short circuit across all the three phases. In majority of cases this current will be the maximum possible current that may flow in the system under a short circuit condition. In some cases the single line to ground fault may exceed this value. Therefore it is reasonably accurate to select/design all the system components/equipment based on this current.

Why do I need to know the short-circuit current?

The maximum short circuit current is calculated for the following reasons:

- To decide on the fault duty of the breakers
- To decide on the size of busbars to be used in a panel and so on.
- To decide on the magnetic trip setting of the breaker.
- For checking the thermal constraint of the cables anticipated to carry the short circuit current.

The CS employs the resistance-reactance method as per IEC-781 for the calculating the short circuit current. The formula for calculating the short circuit current is as given below:

$$Isc = \frac{C \times Vn}{\sqrt{3} \times Z_{kt}} \quad kA$$

Where C = 1 for 230 / 400V; C = 1.05 for 400V < Vn < 1kV; C = 1.1 for Vn > 1kV

The following content of this section will illustrate the method for computing the value of Z_{kt} for different system components.

3.3.1 – Upstream Parameters:

Generally, the electricity supply of the region will furnish the short circuit level of the system at the point of supply to a customer. For MV/LV substation the value is typically around 350MVA. The upstream impedance, resistance and reactance values will then be calculated as mentioned below:

$$Z_{kt} = \frac{Vn^2}{S_{up} \times 1000} \quad m\Omega$$

Note: S_{up} value should be taken in MVA

$$R_{kt} = 0.1 \times Z_{kt} \quad m\Omega$$

$$X_{kt} = 0.995 \times Z_{kt} \quad m\Omega$$

The R_{kt} and X_{kt} thus obtained become the upstream values for the LV system downstream.

3.3.2 – Transformers:

In short the short circuit current at the transformer terminals can be calculated as the reciprocal of percentage impedance multiplied by its full load current. For eg; a 1000kVA transformer, 11kV/400V, U_k = 5% will have a maximum short circuit current of 100/5 = 20times its full load current which is 1443.42 x 20 = 28.87kA.

The more accurate method for calculating the short circuit current is mentioned below:

$$Z_k = \frac{U_z \times Vn^2}{S \times 100} \quad m\Omega$$

$$R_k = \frac{U_r \times Vn^2}{S \times 100} \quad m\Omega$$

$$X_k = \frac{U_x \times Vn^2}{S \times 100} \quad m\Omega$$

The values of U_z, U_r & U_x can be obtained from the transformer test certificate or from the transformer name plate details.

$$R_{kt} = R_k + R_{kt}\ (\text{upstream})\ \ m\Omega$$
$$X_{kt} = X_k + X_{kt}\ (\text{upstream})\ \ m\Omega$$

$$Z_{kt} = \sqrt{(R_{kt}^2 + X_{kt}^2)}\ \ m\Omega$$

3.3.3 – Motors:

How do motors contribute to the short circuit?

When a short circuit occurs at a point in a system, then theoretically, there will not be any current in other parallel circuits and so, will be for the circuits feeding motor loads. The inertia of the load/rotor of the motor will not allow the motor to come to a halt immediately after a short circuit condition and hence the motor starts working as a generator for a very short duration feeding the faulty circuit.

Hence, it becomes a necessity to consider the motor loads as well for short circuit calculation as it may give a substantially high value of current. The transient impedance of the motor loads shall be calculated as mentioned below which shall be used for calculating the short circuit current due to the motor loads:

$$Z'_m = \frac{Vn^2}{S_m \times I_L/I_F}\ \ m\Omega$$

$$R'_m = 0.42 \times Z'_m \qquad m\Omega$$
$$X'_m = 0.91 \times Z'_m \qquad m\Omega$$

3.3.4 – Combined Effect of Transformer & Motor Loads:

The short circuit current calculated from the Z_{kt} obtained as explained in section 3.3.2 will give the short circuit current by the transformer alone. The more accurate and also the maximum value of the short circuit current will be obtained when the faulty circuit is fed from both the transformer and the motor loads. Hence, the resultant short-circuit impedance will be the parallel combination of the motor and transformer impedances.

$$Z'_m = R_m + jX_m$$
$$Z_k = R_k + jX_k$$

Referring back to the fundamentals of AC circuits the parallel combination of Zm and Zk can be obtained by simply adding the admittances.

$$1/Z_{kt} = 1/Zm + 1/Zk$$

Therefore, $Y_{kt} = Y_m + Y_k$

The admittance Y can be expressed as Y = G – jB where G is the conductance and B the susceptance expressed in Siemens (S).

$$Z = R + jX$$

$$Y = 1/Z$$

$$= \frac{1}{R + jX}$$

$$= \frac{1}{R + jX} \times \frac{R - jX}{R - jX}$$

$$= \frac{R}{R^2 + X^2} - \frac{jX}{R^2 + X^2}$$

Therefore,

$$G - jB = \frac{R}{R^2 + X^2} - \frac{jX}{R^2 + X^2}$$

Hence,

$$G = \frac{R}{R^2 + X^2}$$

$$B = \frac{X}{R^2 + X^2}$$

Conversely, it can be proved in the same method as stated above that,

$$R = \frac{G}{G^2 + B^2}$$

$$X = \frac{B}{G^2 + B^2}$$

The values of G_m, B_m, G_k & B_k shall be evaluated as per the formulae stated above.

$$G_{kt} = G_m + G_k$$
$$B_{kt} = B_m + B_k$$

From these values the values of R_{kt}, X_{kt} and subsequently Z_{kt} can be evaluated and shall be used to find the resultant short circuit current.

3.3.5 – Generators:

When the power system is fed by a captive generator, then the following formulae applies for calculating the short circuit impedance. Generally, the sub-transient reactance (X''_d) value will be furnished in the name plate details of the generator as a percentage of the nominal voltage of the generator.

$$X_{kt} = \frac{X''d \times Vn^2}{S \times 100} \quad m\Omega$$

$$R_{kt} = 0.1 \times X_{kt} \quad m\Omega$$

$$Z_{kt} = \sqrt{(R_{kt}^2 + X_{kt}^2)} \quad m\Omega$$

This value shall be used to calculate the short circuit current at the generator terminals. As a matter of fact, for the same kVA rating the short circuit level at the generator terminals will be much lesser than that of the transformer. Hence, when a system is fed from transformer and a standby generator then all the downstream ratings shall be based on the transformer parameters, to be more conservative, when operated separately. If they are intended to run in parallel then the parallel combination of generator & transformer impedances shall be used.

3.3.6 – Cables:

Cables do not form an active component of a power system network but they do contribute to the resistance and reactance of the system and hence affect the short circuit current as well. The resistance and reactance of the cables are calculated as mentioned below:

$$R_k = \frac{\rho \times L}{A \times N} \quad m\Omega$$

Where, ρ = 18.32 for Copper conductors @ 20°C
ρ = 30.34 for Aluminium conductors @ 20°C

$X_k = 0.1 \times L$ for $A < 120mm^2$
$X_k = 0.09 \times L$ for $A >= 120mm^2$

$R_{kt} = R_k + R_{kt}$ (upstream) $m\Omega$
$X_{kt} = X_k + X_{kt}$ (upstream) $m\Omega$

$$Z_{kt} = \sqrt{(R_{kt}^2 + X_{kt}^2)} \quad m\Omega$$

3.4 MAXIMUM EARTH FAULT LOOP CURRENT CALCULATION:

It is the maximum current that flows in a line-fault point-ECC-source loop. The formula used for calculating the maximum earth fault loop current is the same as the maximum 3phase short circuit current. Its only the computation of the impedance that differs. The reason is that the earth fault loop is formed by a phase conductor as well as the ECC & the cable armour (if present).

For transformers and generators the value of the impedance is the same as that for the symmetrical 3phase short circuit current. The impedance calculation only differs for the cables. Following part of the section illustrates the computation of earth fault loop impedance of the cables.

$$R_c = \frac{\rho}{A \times N} \quad m\Omega/m$$

Where, ρ = 18.32 for Copper conductors @ 20°C

ρ = 30.34 for Aluminium conductors @ 20°C

The same formula is used to calculate the resistance of the ECC. If the cable is armoured then the resistance of the armour shall be obtained from the following tables:

Steel wire armour for XLPE multicore cables	-	BS 5467:1997 – Table-H.2
Aluminium wire armour for XLPE single core cables	-	BS 5467:1997 – Table-H.1
Steel wire armour for PVC multicore cables	-	BS 6346:1997 – Table-H.2
Aluminium wire armour for PVC single core cables	-	BS 6346:1997 – Table-H.1

The earth protective conductor is then formed by the parallel combination of the ECC and the armour. For single core cables the armour of all the three phases are bonded together and hence for such cases the effective armour resistance will be the tabulated value divided by 3.

$$R_p = \frac{R_{ecc} \times R_a}{R_{ecc} + R_a} \quad m\Omega/m$$

It is important to bear in mind that reactance is essentially associated with a circuit or a loop, generally formed by two conductors but for our calculation, where the phase conductor and ECC are not of the same value, the average of the line-neutral reactance value of the two sizes shall be considered. The value Xc + Xp shall be obtained from the manufacturer's table or else calculated in accordance with Appendix-1 of IEE Guidance Notes-6.

Note: The value of Xc + Xp is 0.3 for multicore armoured cable irrespective of the size of the conductors as per Clause-6.3.1 of IEE Guidance Notes-6.

Therefore the unit earth fault loop impedance is given by

$$Z_u = \sqrt{((R_c + R_p)^2 + (X_c + X_p)^2)} \quad m\Omega/m$$

The circuit earth fault loop impedance is given by

$$Z_c = Z_u \times L \quad m\Omega$$

The total earth fault loop impedance is given by

$$Z_t = Z_c + Z_{up} \quad m\Omega$$

Hence, the maximum earth fault loop current is given by

$$Ief = \frac{Vn}{\sqrt{3} \times Z_t} \quad kA$$

3.5 LET THROUGH ENERGY / THERMAL CONSTRAINT:

The short circuit current while passing through a cable tend to increase the temperature of the conductor considerably and hence it is extremely important to check the adequacy of the cable to withstand the thermal energy being dissipated before the protective device actually trips. The 3phase symmetrical short circuit current being the highest current to be carried by the phase conductor is adopted in sizing the phase conductor and the earth fault loop current is adopted for sizing the ECC.

The adiabatic equation (Clause 543-01-03 IEE Wiring Regulation) gives the minimum required size of a cable to carry a fault current of a given magnitude for a given duration.

$$S_1 = \frac{Isc\sqrt{t_1}}{k_1} \quad mm^2$$

$$S_2 = \frac{Ief\sqrt{t_2}}{k_2} \quad mm^2$$

The value of ‘t’ shall be obtained from table 54C of IEE wiring regulations and the time ‘t’ shall be obtained from the tripping curve of the breaker.

3.5 MAXIMUM EARTH FAULT LOOP IMPEDANCE:

The maximum earth fault loop impedance is calculated to ensure that the earth fault loop impedance value is low enough to trip the protective device within the time permitted in the IEE Wiring regulations (typically 0.4s) in case of an Earth Fault.

The calculation of the circuit impedance follows the same method as for the Earth fault loop current calculation with appropriate correction factors to take into account the maximum temperature attained in case of fault.

The following factors shall apply to the resistance values obtained in the Earth Fault Loop Current calculation for the initial and final temperatures specified in 54C of IEE wiring regulations during a fault:

PVC	-	1.38
XLPE	-	1.60

For armour resistance the following factors shall apply

PVC/SWA	-	1.19
XLPE/SWA	-	1.28
PVC/AWA	-	1.19
XLPE/AWA	-	1.28

An additional correction factor of 1.1 shall be applied to the resistance of steel wire armour to account for the magnetic effect of the armour.

The reactance values remain unaffected with the temperature.

The maximum value of earth fault loop impedance allowed by a protective device follows from its magnetic trip current threshold. It shall be calculated as given below:

$$Zm = \frac{Vn}{\sqrt{3} \times Im} \quad m\Omega$$

Therefore, the cable shall be selected such that the circuit impedance

Z'c < Zm

4 GETTING STARTED WITH CS:

The first and foremost step in starting with a calculation is to input the project details in the title sheet bearing the Tab Name "Ti". The project voltage, frequency, ambient conditions, types of cables/wires for lighting, small power and HVAC, allowable voltage drop percentage at each level shall be furnished in the Title sheet.

4.1 PREPARATION OF DISTRIBUTION BOARD SCHEDULES:

The actual calculation starts with the distribution boards (DB). All the circuit details shall be entered in the sheet bearing the title "Distribution Board Calculation". The next sheet will generate the Load Schedule for the DB in the format as required by DEWA. The following steps will guide through the procedure for carrying out the DB calculations.

- Enter the DB reference, panel type, location and mounting type in the corresponding cells on the top of the sheet.
- Enter the load description under the column (L).
- Enter the no. of units of different type of loads in the columns (M) to (Y).
- Enter no. of phase in the column (AH).
- For lighting loads enter the type of control employed under the column (J) and the device used to control in the column (K).

- Enter the code corresponding to the wire/cable selected for the circuit, from the Title sheet, under the column (E)
- Enter the length of the circuit under the column (AR).
- Enter the load code in the very first column (A) of the sheet which is outside the print area. Enter the correct code for each of the loads. For help regarding the code refer column (BI) and corresponding description in the column (BJ).
- Enter the demand factors in the column (BN) for each type of loads.
- Check the next tab bearing the same name as the DB sheet but with a suffix 'a' and verify if all the circuits entered in the DB sheet is duplicated in this sheet. Insert or delete the rows to match the DB sheet.
- If the user needs to add extra rows then copy the formula of the entire row above and paste them in the new rows until the last row.

Convention: The names appearing in parenthesis refer the column reference as in MS Excel and the ones in quotes refer to the title of the column.

Tips: Do not try to balance the loads the while entering the circuit description. It is advisable to balance the loads once all the circuits are entered to save time in frequent rearrangement. Further do not select the wire size in the beginning because initially the values of upstream voltage will not be available. Do not panic at seeing a lot of errors on the sheet. They will vanish once the entire calculation is done.

Warning: ***The DB sheet and its output sheet should always be copied in pair to keep the links as such. Otherwise it becomes a tedious process to link the sheets. Also, do not alter anything appearing outside the print area except for the load codes in the column (A) and demand factor in column (BN). The load codes shall be correctly entered to obtain the exact classification of loads (lighting, power, HVAC, etc.)***

4.2 PREPARATION OF SUB-MAIN DISTRIBUTION BOARD SCHEDULES:

The next step will be the preparation of SMDB schedules. There are only a very few user inputs with regards to the SMDB schedules. The following steps will guide through the procedures for preparing the SMDB schedules:

- Enter the SMDB reference in the corresponding cell.
- Enter the tab names of the DB's in the column (A) in order of user's preference or as required. The connected load for the DB entered will appear in the corresponding cells. All other values will be displayed once the entire calculation is done.
- If there are loads other than the DB's which are directly fed from the SMDB, then the description shall be entered manually in the column (B) and the loads in the columns (M), (N) & (O). Following that, the load code for such loads should be entered in the column (V).
- After entering all the loads press the 'Calculate' button on the left-bottom of the sheet. It will take sometime to process the calculation and once it is done the demand load will be displayed. The connected and demand load should be the same as in the cells (BL17) and (BP17), otherwise it will be displayed in red colour. For such cases check if the load codes are entered for all the final circuits in SMDB and also in the DB's fed from the SMDB
- Enter the demand factors in the column (BN) for each type of loads.

It is advisable to name the spares uniquely in all the SMDB's, MCC's, etc. such as 'Spare' suffixed by the SMDB name or so, so as to have the correct look-up value for each spare way when referred from the Calc sheet.

4.3 PREPARATION OF MOTOR CONTROL CENTRE SCHEDULES:

The next step will be the preparation of MCC schedules (if any). The following steps will guide through the procedures for preparing the SMDB schedules:

- Enter the MCC reference in the corresponding cell.
- All the load description shall be entered manually in the column (B) and the loads in the columns (M), (N) & (O). Following that, the load code for such loads should be entered in the column (V).
- After entering all the loads press the 'Calculate' button on the left-bottom of the sheet. It will take sometime to process the calculation and once it is done the demand load will be displayed. The connected and demand load should be the same as in the cells (BL17) and (BP17), otherwise it will be displayed in red colour. For such cases check if the load codes are entered for all the final circuits.
- Enter the demand factors in the column (BN) for each type of loads.

4.4 PREPARATION OF BUS DUCT SCHEDULES:

The preparation of Bus Duct schedules follows the same procedure as for the SMDB or MCC schedules. In addition to that the user has to fill in the values in the columns (F) & (G) starting from row (41) to (47).

4.5 PREPARATION OF MDB & TRANSFORMER SCHEDULES:

The preparation of MDB and Transformer schedules follow the same steps as for the SMDB or MCC schedules.

4.6 WORKING WITH CS:

The CS forms the master calculation sheet for any project. Its only after completing this sheet that the entire output is generated in other sheets. The procedure for working with CS is as illustrated in the following steps but not limited to those steps only. More often, it would require user input based on the Check remarks. As a general rule, all the cells highlighted in yellow color require user input.

- Enter/Check the design conditions on the top right corner of the sheet.
- Complete the transformer details on the top-centre of the sheet by linking the kVA rating cell to the corresponding cell in the Transformer Load Schedule sheet.
- The first row of the sheet after the title rows is for the upstream network. Enter "Upstream Network" under the column (F) and then the upstream short circuit level, No. of phases & power factor in the corresponding cells. No other data apply for the upstream network.

- The second row is for the short circuit current calculation at transformer secondary terminal. Enter "Upstream Network" under the column (E) and Transformer reference starting with "Tx-"followed by a number under the column (F). Enter the no. of phases under the column (H). This should give the short circuit current at the transformer secondary automatically.

Warning: ***The word "Upstream Network" and the Transformer reference should be the same everywhere throughout the workbook. The transformer reference should always be referred as "Tx-" followed by a number (1,2,3 or so on) for correct working of the sheet.***

- The third row is for the feeder from the Transformer to the Main LV Panel or MDB. Enter "md" in the column (C). Enter the main LV panel reference fed from the subject transformer with a suffix "LESS MOTOR" to indicate that the short circuit current is calculated without motor contribution. Link the cell under the column (J) with demand load cell in the corresponding LV panel load schedule. Enter the load power factor under the column (L). Enter the protective device setting value, select the protective device frame rating and magnetic current setting multiplier value in the columns (N), (O) & (P) respectively. Select the installation method, installation method description, enter busbar rating (if applicable), feeder length and conductor insulation under the columns (AS), (AT), (AU), (AV) & (AW) respectively. Enter the no. of circuits for grouping, spacing, Cable size and No. of parallel feeders, no. of parallel neutral conductors (if different from phase conductors) under the columns (BC), (BD), (BE), (BF) & (BG) respectively. All the other values will pop up automatically. If there is any remark in the 'Check Remarks' column then try changing the cable size or as required until the warning in RED FONTS disappear.

Tips: For cable size entry, it is advisable to start with the DEWA recommended size and then modify the size as required by the calculation.

- The fourth row is for the short circuit current calculation considering the transformer and motor loads together. Enter "md" in the column (C). Enter "MDB-1 (INCLUDING MOTOR)" in the column (F). Enter no. of phases in the column (H). Enter the total motor load to be considered for short circuit calculation in the column (AI). Enter the ratio of locked rotor current to the full load current in the column (AH). The maximum short circuit current value will pop-up automatically in the respective cell.

Since, we have two different rows relating to the Main LV panel (3rd and the 4th row) it has to be summarized in a single row bearing all the final parameters to be considered downstream.

- So, the fifth row is same as the third row except that the cells under the short circuit calculations are linked to the corresponding cells in the 4th row. Check for any remarks under the 'Check Remarks' column and modify the cable size or as required by the remarks.

For almost all the LV calculations the first 5rows shall be the same except for few values like transformer rating, load kW, etc.

The following steps will include the distribution starting from the Main LV Panel till the last DB. These steps are going to be the generic ones.

- Enter the source tab name under the column (B). Source tab name is the tab name of the sheet having the source load schedule.
- Enter the type of panel in the column (C). The code identifies whether the source is a MDB or SMDB or MCC or Busbar or DB or motor load or final circuit. This should display the Source point, Network reference point, load kW, breaker details etc.
- If the Network reference point is a Power Factor Correction Capacitor Bank enter the reactive power rating in the column (K).
- Enter the No. of phases in the column (H).
- Enter the load power factor in the column (L).
- Enter the protective device setting, protective device frame rating and magnetic current trip setting multiplier in the columns (N), (O) & (P) respectively.
- If the Network reference point is a motor load then, enter the type of starter and rating of earth leakage current setting in the columns (U) & (W).
- Select the installation method, installation method description, enter busbar rating (if applicable), feeder length and conductor insulation under the columns (AS), (AT), (AU), (AV) & (AW) respectively.
- Enter the no. of circuits for grouping, spacing, Cable size and No. of parallel feeders, no. of parallel neutral conductors (if different from phase conductors) under the columns (BC), (BD), (BE), (BF) & (BG) respectively.

All the other values will pop up automatically. If there is any remark in the 'Check Remarks' column then try changing the cable size or as required until the warning in RED FONTS disappear.

The disconnection time for the protective device setting shall be verified for the magnetic trip threshold from the tripping curve of the protective device being used. The value "0.05s" holds good if the short circuit current at the network reference point falls in the reflex tripping zone of the breaker tripping curve. Therefore, it is advisable to check the tripping curve of the selected protective device.

Follow the above steps for the next feeder till all the feeders are included. As a good practice, while starting with an MDB distribution, when a SMDB/MCC/Busbar is encountered which is expected to have further distribution downstream, it is advisable though not mandatory to include all the feeders emanating from them unit the final DB or load before moving on to the next feeder.

When it is required to insert rows, somewhere in the middle, the user has to copy the cell under the column (A) from the row above and paste them in the corresponding cell in the new row & the row below it in order to maintain the continuity of the row numbering.

Although the CS gets the value of Source Point & Network Reference Point automatically, in some cases it may be required to manually enter few feeder details, for e.g. the feeder for the UPS bypass supply to the UPS board would require manual inputs in those cells and also in the Load cell.

Having done all the calculations in the CS it now becomes important to check if all the values are correctly appearing in all the load schedules. The final step is to select the wire size in the Distribution board calculations and then verify if all the values are correctly reflected in the output sheet. It is advisable to the user to double check all the

sheets thoroughly including the cells having formulae before issuance. This completes the calculation of a project.

5 LICENSE AND DISCLAIMER:

The CS as well as the information contained in the user manual has been developed in close conformance to the 16th Edition of IEE Wiring Regulations: 2000 and results are found to be in accordance with the manual calculations. Sufficient measures have been taken to be on the most conservative and economical side possible. However, neither the developer nor the approver guarantees on the results generated by the CS. It is the sole responsibility of the user to familiarize himself/herself with the concepts and verify the results on his/her own.

The CS being the first version of its kind may have errors. Therefore, please do not hesitate to suggest us of any corrections or improvements to be done on the CS.

The CS becomes the proprietary spreadsheet of the developer and W.S.Atkins and Partners Overseas and so is the information contained in this manual and is available for use by W.S.Atkins & Partners Overseas staff only.

6 ABOUT CABLESIZER VERSION 1.1:

6.1 CABLESIZER VERSION 1.1

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6.2 CABLESIZER VERSION 1.1 USER MANUAL

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APPENDICES

7.1 - IEC 364-5-523 SCHEDULE OF INSTALLATION METHODS

7.2 - CURRENT CARRYING CAPACITY

TABLES – DUCAB

7.3 – DERATING FACTORS – DUCAB

7.4 - VOLTAGE DROP TABLES, IEE WIRING REGULATIONS EXTRACTS

7.5 - RESISTANCE & REACTANCE VALUES – RIYADH CABLES, BS 5467:1997 & BS 6346:1997 EXTRACTS, IEE WIRING REGULATIONS EXTRACTS, IEE GUIDANCE NOTES-6 EXTRACTS

7.6 – BUSBAR TECHNICAL DATA -

POWERBAR

7.7 – APPENDIX – 4 – IEE WIRING REGULATIONS

7.8 – MERLIN GERIN BREAKER CATALOGUE

EXTRACT & TRIPPING CURVES

7.9 – TYPICAL TRANSFORMER TEST CERTIFICATE